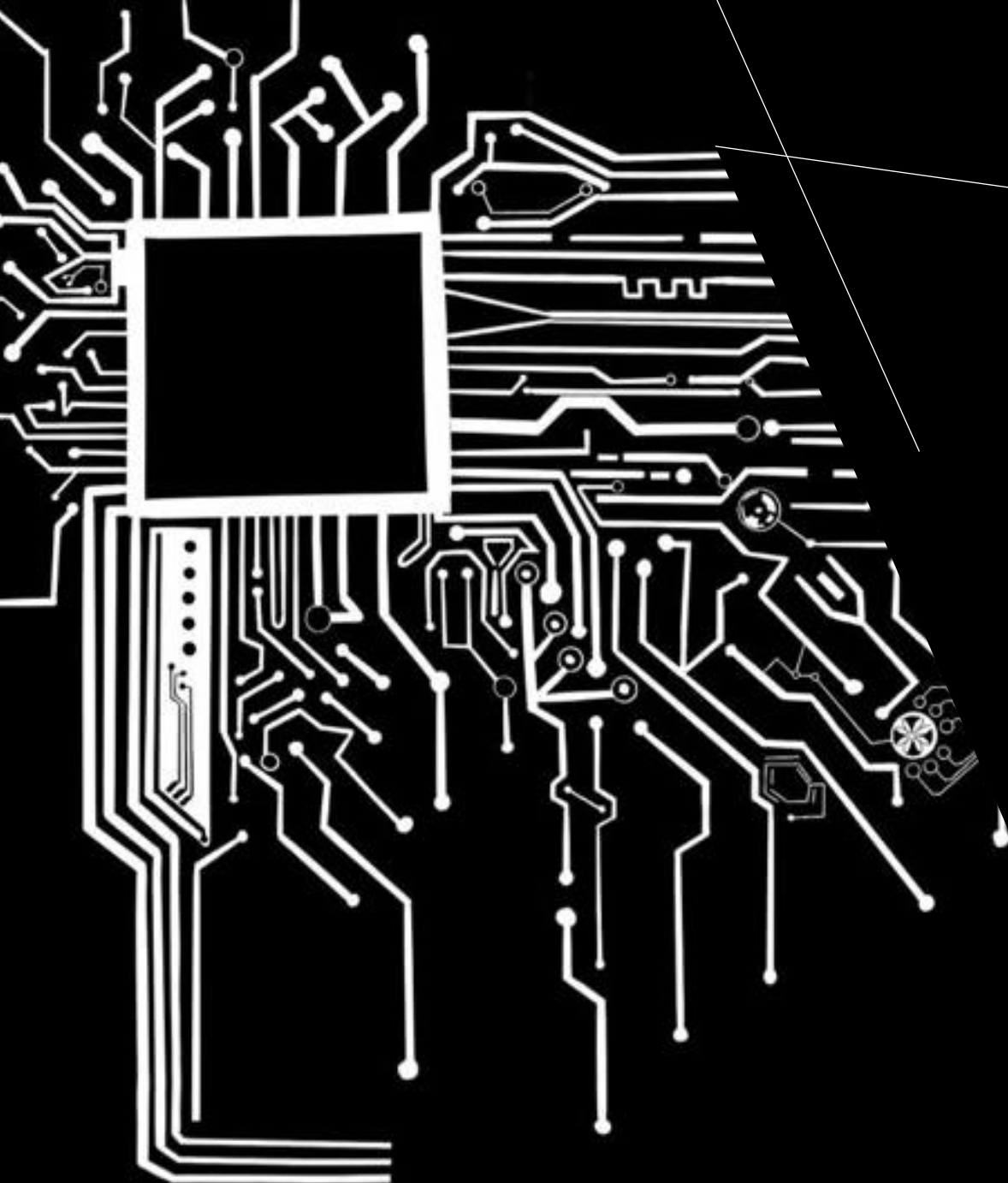




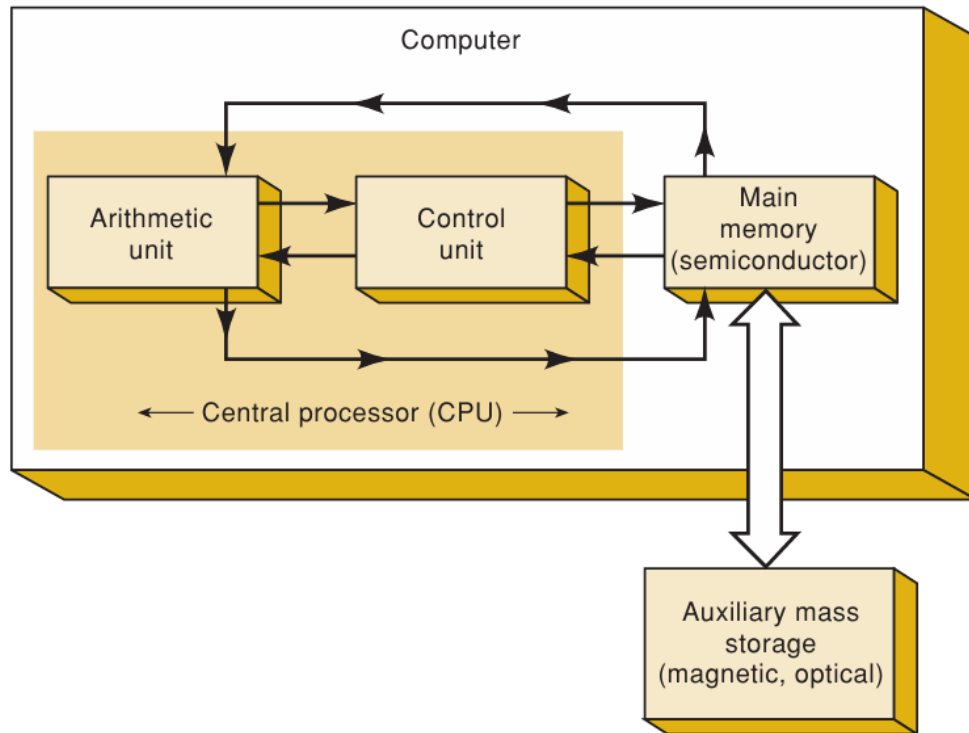
ICE 2343

DIGITAL LOGIC DESIGN

ABHIRUP SARKER
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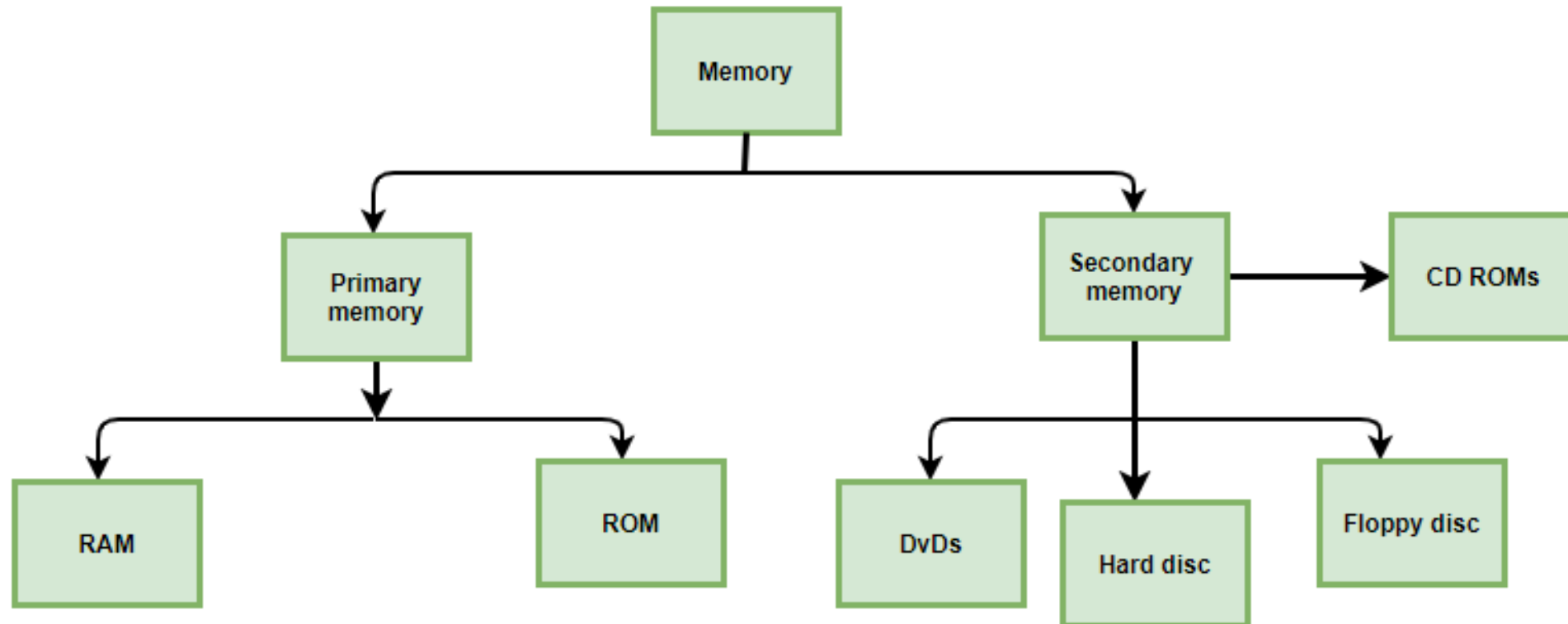
Memory Devices

A major advantage of digital systems over analog systems is the ability to easily store large quantities of digital information and data. For example, in a digital computer, the internal main memory stores instructions that tell the computer what to do under all possible circumstances so that the computer will do its job with a minimum amount of human intervention.



- **Main memory:** Semiconductor memories are used as main memory devices. A computer's main memory is also called its **working memory or primary memory**. It is in constant communication with the central processing unit (CPU) as a program of instructions is being executed. Example: RAM, ROM
- **Auxiliary memory:** It is separate from main working memory, also known as **storage memory or secondary memory**. It stores data that is currently not being used by the CPU. It can store massive amount of data but comparatively slower than main memory. Example: SSD, HDD, CD etc.

Classification



Random Access Memory

RAM is volatile memory which stores data temporarily, meaning when power turned off the data is erased. RAM stores the data that is being used during the execution of a program.

Characteristics:

- High speed performance
- Volatile nature
- Multitasking
- System program and instruction execution

Two types of RAM:

1. Static Random Access Memory (SRAM): Static RAM stores data using flip-flop circuits and retains information as long as power is supplied without refresh.
2. Dynamic Random Access Memory (DRAM): Dynamic RAM stores data as charge in capacitors that must be periodically refreshed to maintain information.

SRAM vs DRAM

SRAM	DRAM
It consists of flip-flops for storing data	It is made of capacitors that store data
It consumes more power than DRAM	It is power efficient compared to SRAM
SRAM used in cache memory for quick access	DRAM used in main memory for large data storage
It is faster	It is slower
More expensive	Less expensive

Pros and Cons of RAM

Advantages:

- RAM is faster than any other type of storage.
- RAM allows a computer to handle multiple applications simultaneously without slowing down.
- CPU can read and write data directly from RAM during program execution.

Disadvantages:

- Data is lost when power turned off.
- RAM can be very expensive.
- Needs power to retain information.
- It requires large physical space.

Read Only Memory

RAM is non-volatile memory; it stores data permanently and retains it even when power is turned off. ROM has decoder and OR gates that processes the data and store in registers.

Advantages:

- Retains data even when power is off.
- Data remains secure and doesn't accidentally get erased.
- Low production cost.
- Quick data access.

Disadvantages:

- Data can't be easily modified.
- Slow writing compared to RAM.
- Not suitable for large amount of data storage.

Types of ROM

- **Masked ROM (MROM):** MROM has its information stored at the time the integrated circuit is manufactured. It is made of rectangular blocks of transistors. MROM can not be reprogrammed by the user.
- **Programmable ROM (PROM):** PROM are manufactured blanked and can be programmed by the user. Once programmed, a PROM is like an MROM because it cannot be erased and reprogrammed. It has fusible-link PROM structure, these fuses store data as 0/1.
- **Erasable Programmable ROM (EPROM):** An EPROM can be programmed by the user, and it can also be erased and reprogrammed as often as desired. The storage element of an EPROM is a MOS transistor and can be erased using ultraviolet light.
- **Electrically Erasable Programmable ROM (EEPROM):** EEPROM can be electrically erased and reprogrammed as many times. Its structure is similar to EPROM but more compact and improved performance. High voltage signal charges the floating MOS transistors that store the data.

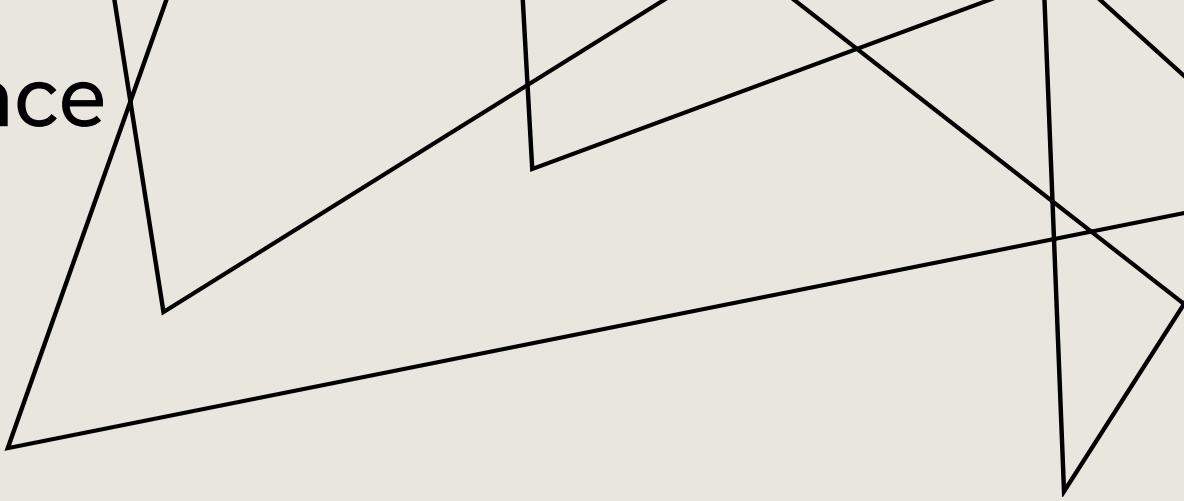
RAM vs ROM

RAM	ROM
Data can be easily be modified	Data can not be modified
RAM is faster than ROM	ROM is slower than RAM
RAM is volatile	ROM is non-volatile
RAM is expensive	ROM comparatively cheaper
Used for storing data currently processed by CPU	Used to store firmware and BIOS data
Enables read and write operation	Allows only read operation

Other Memory Devices

- **Cache Memory:** Cache memory is a high-speed semiconductor memory placed between the CPU and main memory. It stores copies of frequently accessed instructions and data so the processor can retrieve them with very low latency. Cache memory is very small size and fast processing. It is volatile like SRAM.
- **Hard Disk Drive (HDD):** It is a secondary storage used for storing large information. HDD is a magnetic secondary storage device, non-volatile, stores backups, archives and operating systems programs.
- **Solid State Drive (SSD):** SSD is a flash-memory-based secondary storage device that stores data electronically instead of magnetically like HDD. SSD is faster than HDD. The data is stored as electrical charge in memory cells. It is more silent, durable, doesn't have movable parts like HDD but expensive.

Reference



Digital Systems Principles and Applications by Ronald J.Tocci
Chapter- 12

A series of white, thin, overlapping geometric lines on a black background, forming a complex, abstract pattern on the left side of the slide. The lines intersect to create various polygonal shapes, some of which are nested within others.

THANK YOU